

A: Account Qualifying

Source file name: `accountq.c`, `accountq.cpp`, `accountq.java`, or `accountq.py`

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The Bank of Linearonia (BoL) is rolling out a qualification process for customer accounts. For each account, the bank will compute a set of indices to rate its behavior over a given period, e.g., maximum, average, and minimum values, as well as the number and types of transactions.

Suppose an account X has N transactions $X[0], X[1], \dots, X[N-1]$. A *transaction* is an integer: a *deposit* if it is positive, a *balance inquiry* if it is zero, and a *withdrawal* if it is negative. BoL wishes to compute the following indices:

d : the *maximum deposit*;

w : the *maximum withdrawal*; and

r : the *longest paired subperiod*, understood as the length of the longest subperiod that contains as many deposits as withdrawals.

If there are not deposits, the index d is 0. If there are not withdrawals, the index w is 0. However, if there are withdrawals, w is the most negative of them.

For instance, consider an account with transactions 23, -12, 0, 15, 0, 5. In this case, the indices are:

$$d = 23, \qquad w = -12, \qquad r = 4.$$

The longest paired subperiod for this example is $r = 4$ because the subperiod $X[1..4]$ has length 4 and contains the same number of deposits and withdrawals (zeros do not affect the counts). No other subperiod has length greater than 4 with equal numbers of deposits and withdrawals. Note that care is required: for the sequence $(-3, 0)$ the longest paired subperiod is $r = 1$, since the subperiod consisting of the single element 0 has 0 deposits and 0 withdrawals.

Your task is to write a program that assists BoL in the account qualification process.

Input

The input consists of multiple test cases. Each test case begins with a line containing an integer N ($0 < N \leq 10\,000$), the length of the period. The next line contains exactly N integers $X[0], X[1], \dots, X[N-1]$ separated by single spaces, where $-10\,000 \leq X[i] \leq 10\,000$ corresponding to a sequence of transactions X . A line containing a single 0 marks the end of input and is not a test case.

The input must be read from standard input.

Output

For each test case, output one line with the three blank-separated integers d , w , and r corresponding to X .

The output must be written to standard output.

Sample Input	Sample Output
6 23 -12 0 15 0 5 2 -3 0 3 100 200 300 3 0 0 0 0	23 -12 4 0 -3 1 300 0 0 0 0 3